PART 1 - CAPSTONE PROJECT #6

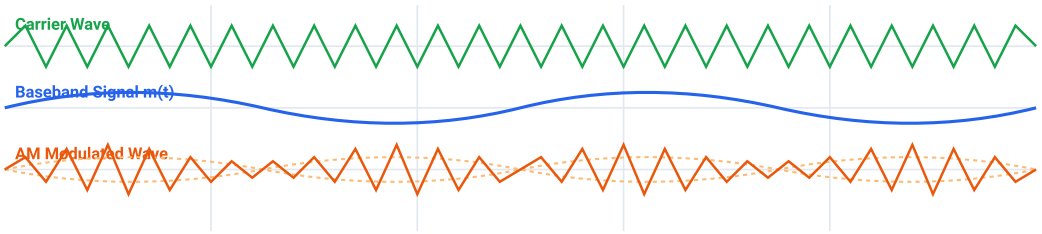
Modulation Fundamentals & Theory

Tasks 61–64: Beginner and Advanced Explanations with Verified Mathematical Equations.

1. Analog Modulation Systems (AM & FM)

1.1 Amplitude Modulation (AM)

Beginner Explanation: In AM, the amplitude (height) of a high-frequency carrier signal changes in direct proportion to the message signal, while frequency and phase remain constant.



$$s(t) = A_c\,[1 + \mu \cdot \cos(2\pi f_m t)] \cdot \cos(2\pi f_c t)$$

Modulation Index: $\mu = A_m / A_c = (A_{max} - A_{min}) / (A_{max} + A_{min})$

Bandwidth: $B_{AM} = 2 \cdot f_m$

1.2 Frequency Modulation (FM)

Beginner Explanation: In FM, peak amplitude remains fixed, but frequency changes according to message signal amplitude.

$$s(t) = A_c \cdot \cos[2\pi f_c t + \beta \cdot \sin(2\pi f_m t)]$$

Modulation Index: $\beta = \Delta f / f_m$ | **Carson's Bandwidth:** $B_{FM} = 2(\Delta f + f_m)$

2. Digital Keying Systems (ASK & FSK)

2.1 Amplitude Shift Keying (ASK)

Beginner Explanation: Binary '1' turns carrier wave ON, while binary '0' turns it OFF (On-Off Keying).

$$s_{ASK}(t) = d(t) \cdot A_c \cdot \cos(2\pi f_c t),\ d(t) \in \{0, 1\} \quad | \quad \textbf{Bandwidth:}\ B_{ASK} \approx R_b$$

2.2 Frequency Shift Keying (FSK)

Beginner Explanation: Binary '1' uses high frequency f_1 and binary '0' uses low frequency f_0.

$$s_{FSK}(t) = A_c \cdot \cos(2\pi f_i t),\ f_i \in \{f_1, f_0\} \quad | \quad \textbf{Bandwidth:}\ B_{FSK} = |f_1 - f_0| + 2 R_b$$

3. Comparative Matrix

Type	Parameter Varied	Noise Immunity	Bandwidth
AM	Carrier Amplitude	Low	$2 f_m$
FM	Carrier Frequency	High	$2(\Delta f + f_m)$
ASK	Digital Amplitude	Low	$\sim R_b$
FSK	Digital Frequency	Moderate	$\lvert f_1 - f_0 \rvert + 2R_b$